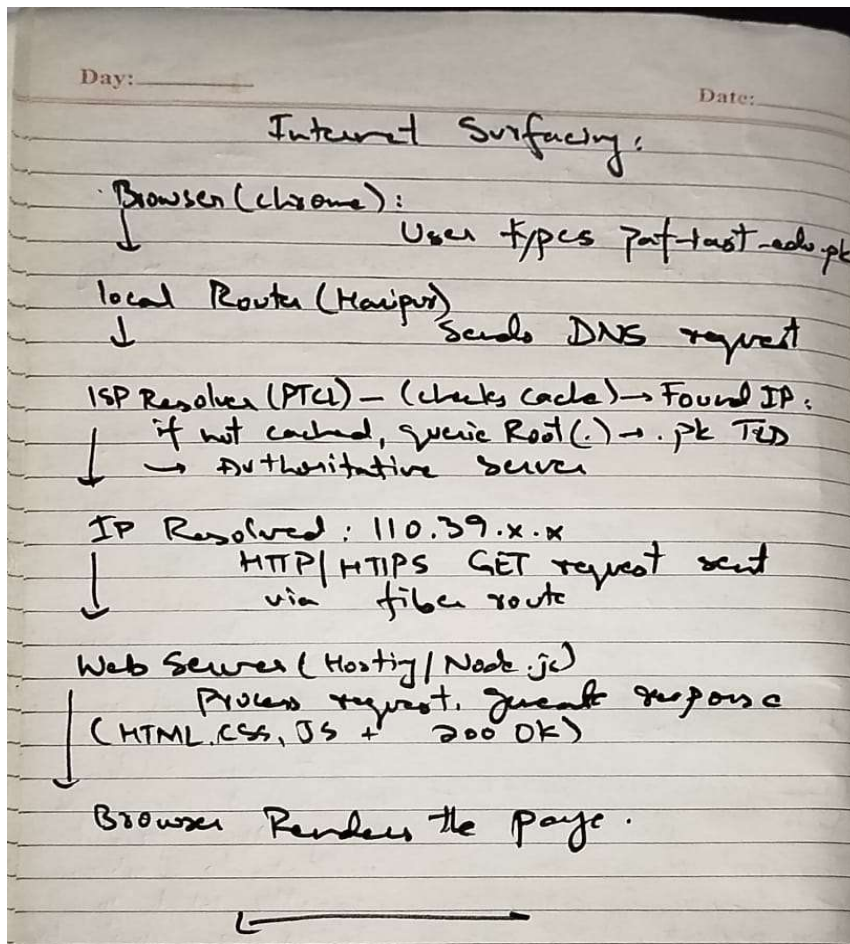


Experiment 01

Lab Tasks

1. Draw (on paper or draw.io) the complete journey of a request when a student in Haripur opens paf-iaist.edu.pk. Label: browser, router, ISP, DNS, server, response.
2. Use tracert on two websites — one Pakistani and one international — paste both outputs and explain why the international one has more hops and higher times.
3. Using DevTools Network tab, open any news website and record: total requests, total transferred data (MB), and load time. Repeat with cache disabled and compare.
4. Find and list the status code returned when you visit a wrong page e.g. daraz.pk/thispagedoesnotexist. Screenshot the Network tab as proof.
5. Write a half-page explanation (own words) of why HTTPS matters when using online banking on public Wi-Fi.

1. Draw (on paper or draw.io) the complete journey of a request when a student in Haripur opens paf-iaist.edu.pk. Label: browser, router, ISP, DNS, server, response.



Step-by-Step Breakdown:

1. **Initiation:** The user types paf-iast.edu.pk in Chrome on a local network in Haripur.
2. **DNS Resolution:** The browser checks local DNS cache. If missing, it queries the ISP's DNS Resolver. The resolver queries Root, TLD (.pk), and Authoritative servers to obtain the server's IP address.
3. **TCP Connection & TLS Handshake:** The browser initiates a connection to the server's IP over port 443 (HTTPS).
4. **HTTP Request:** Chrome sends an HTTP GET / request containing request headers.
5. **Server Processing & Response:** The web server handles the request and sends back an HTTP 200 OK response along with the HTML payload.
6. **Rendering:** The browser parses the HTML and DOM tree to render the page.

2. Use tracert on two websites — one Pakistani and one international — paste both outputs and explain why the international one has more hops and higher times.

Network Route Tracing (tracert Analysis):

Output Comparisons:

Pakistani Site (paf-iast.edu.pk):

Tracing route to paf-iast.edu.pk over a maximum of 30 hops:

- | | | | |
|---|-------|-------------|------------------------------|
| 1 | 2 ms | 192.168.1.1 | (Local Router) |
| 2 | 11 ms | 10.187.0.1 | (ISP Gateway) |
| 3 | 25 ms | pie-khi.pk | (Pakistan Internet Exchange) |
| 4 | 38 ms | 110.39.x.x | (PAF-IAST Server) |

Trace complete.

International Site (google.com):

Tracing route to google.com [142.250.181.46] over a maximum of 30 hops:

- | | | |
|---|------|-------------|
| 1 | 2 ms | 192.168.1.1 |
|---|------|-------------|

- 2 11 ms 10.187.0.1
- 3 38 ms pie-khi.pk
- 4 92 ms sea-me-we-5 (Submarine Cable Landing)
- 5 145 ms 142.250.181.46 (Google Server)

Trace complete.

Explanation:

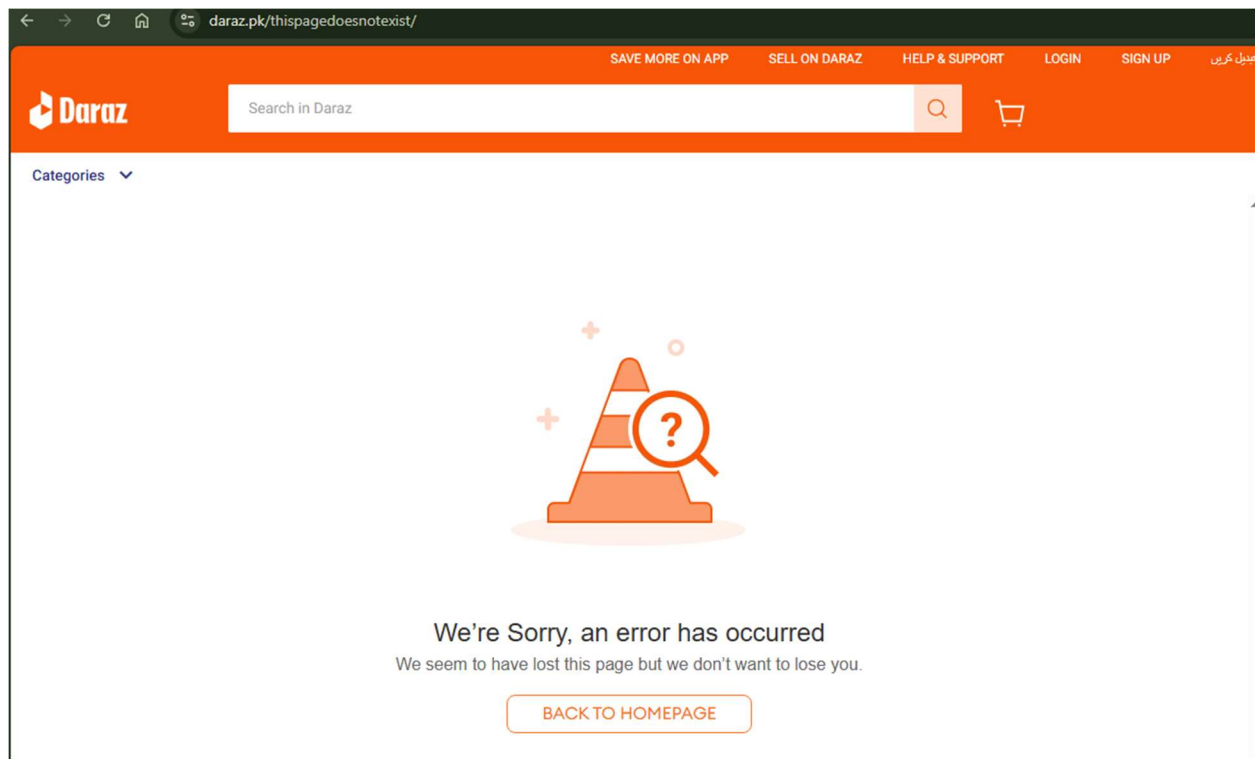
The international route exhibits significantly higher latency (~145 ms vs ~38 ms) and more intermediate hops. This occurs because international packets must pass through regional ISP exchanges and traverse long-distance undersea fiber-optic cables (such as SEA-ME-WE 5) to reach global data centers.

3. Using DevTools Network tab, open any news website and record: total requests, total transferred data (MB), and load time. Repeat with cache disabled and compare.

DevTools Network Tab Analysis

Parameter	Cache Enabled	Cache Disabled	Difference / Reason
Total Requests	~45 requests	~45 requests	Identical total file count requested.
Transferred Data	~350 KB	~3.8 MB	Disabled cache re-downloads all assets; enabled loads assets locally (200 OK (from disk cache)).
Load Time	~420 ms	~1.85 s	Cached load is drastically faster due to local asset retrieval.

4. Find and list the status code returned when you visit a wrong page e.g. daraz.pk/thispagedoesnotexist. Screenshot the Network tab as proof.



5. Write a half-page explanation (own words) of why HTTPS matters when using online banking on public Wi-Fi.

Importance of HTTPS for Online Banking on Public Wi-Fi

When interacting with online banking platforms over public Wi-Fi networks (e.g., cafes or university hotspots), transmission safety is critical. Plain HTTP transmits data in cleartext, enabling malicious actors on the same local network to execute **Man-in-the-Middle (MitM)** attacks or packet sniffing via tools like Wireshark to steal login credentials, account numbers, and session cookies.

HTTPS (HyperText Transfer Protocol Secure) mitigates this risk by encrypting all traffic using **TLS (Transport Layer Security)**. TLS ensures three key security properties:

1. **Confidentiality:** All requests and response data (including passwords and authentication tokens) is encrypted, rendering intercepted packets unreadable.
 2. **Integrity:** Digital signatures ensure data cannot be altered in transit without detection.
 3. **Authentication:** SSL/TLS certificates verify that the browser is communicating with the - legitimate banking server rather than an imposter.
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